Chapter 4

Welding

Visit for more Learning Resources

CO

- Identify joining methods for fabrications, and fabricate simple job using joining method.

Welding

Welding is metallurgical fusion process,where part to be joined by the application of heat and pressure.

- It is used to join the two different parts.
- It is commonly used in the industries.
- ## Advantages.
- It is used for repair work.
- Strength of joint is good.
- They have high corrosion resistance.
- Welding is used for fluid tight joints.
- Different types of joints can be done.
- Welding is economical and cost efficient.

Two Categories of Welding Processes

- Fusion welding - coalescence is accomplished by melting the two parts to be joined, in some cases adding filler metal to the joint
  - Examples: arc welding, resistance spot welding, oxyfuel gas welding
- Solid state welding - heat and/or pressure are used to achieve coalescence, but no melting of base metals occurs and no filler metal is added
  - Examples: forge welding, diffusion welding, friction welding

Welding

- **Disadvantages.**
- It produce harmful radiations.
- It produce internal stresses distortion and change in micro structure in welding r egion.
- Previous badge preparation requires.
- Require skilled operator.
- **Applications.**
- Automobile parts.
- Aircraft and ship construction.
- Machine frames.
- Pipelines.
- Tanks and vessels.

Welding classifications

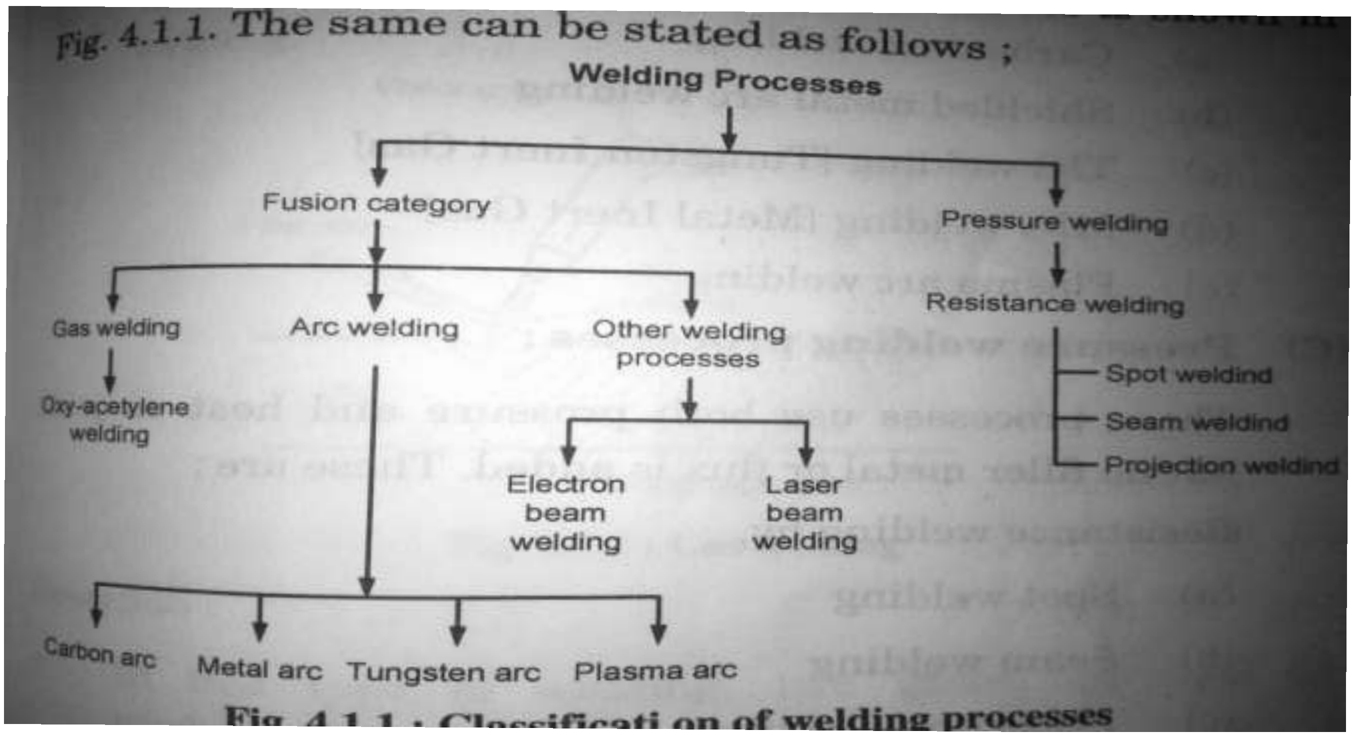


Fig. 4.1.1. The same can be stated as follows ;

Fig. 4.1.1 : Classificati on of welding processes

Welding Classification

- Gas arc welding.

1. Air acetylene welding.
2. Oxyacetylene welding.
3. Pressure gas welding.

- Arc welding.

1. Carbon arc welding.
2. Shielded metal arc welding.
3. TIG welding.
4. MIG welding.
5. Plasma arc welding

Welding classification

- Pressure welding process
- Process requires both heat and pressure for joining. It does not require filler metal.

1. Spot welding.
2. Seam welding.
3. Projection welding.

- Other fusion process.

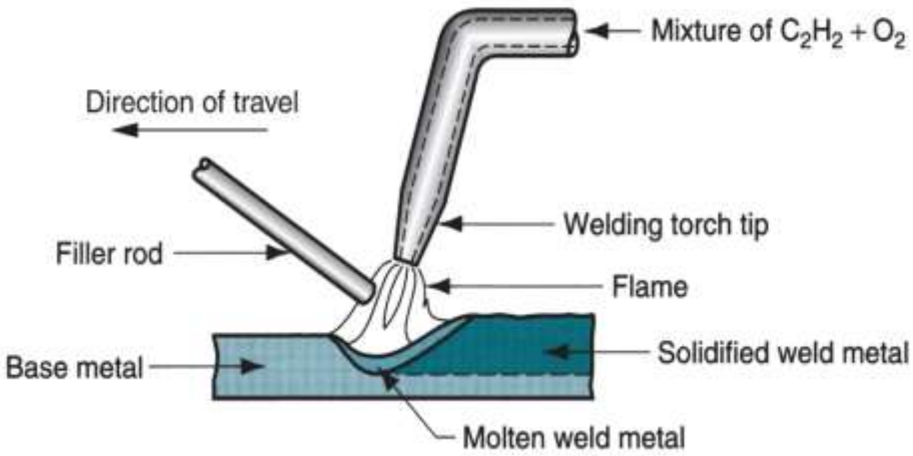
1. Electron beam welding.
2. Laser beam welding.

Acetylene gas welding

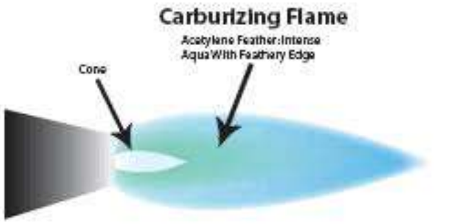
- In this process oxygen and acetylene gas is used for welding process.
- It rises the temperature of metal above melting point.
- Filler material is used for joining.
- Operation
- In this process oxygen and acetylene is used for the welding.
- Oxygen supports the combustion and acetylene gas generate high temp.
- Operation starts with placing the part togather which is to be join.
- Oxiacetylene flame heat the metal temp produced is about 3200°c.
- Filler metal is added to the join which join the parts by melting it self.
- filler material is copper coated steel, carbon steel.
- chemical reaction at tip is as followed.
- $2C^2H^2+5O^2 \rightarrow 4CO^2+2H^2O$

WELDING PROCESSES

- Advantages.
- It is used for all types of joints.
- Oxyacetylene flame can be easily controlled.
- Suitable for thin metal sheets.
- Equipment is portable & versatile.
- Limitations.
- Method is slower.
- Distortion of workpiece is more.

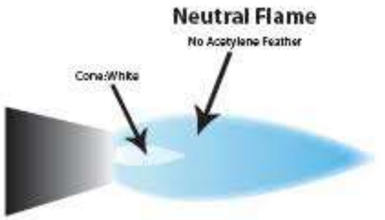


Types of flames



- Natural flame.

1. All gas is burn
2. gives 100% heat.
3. Ideal flame used in gas welding.



- Reducing flame.

1. It consist of inner core intermediate feather and outer flame.
2. It is also called as carburising flame.
3. strong reduction in higher zone due to high amount of acetylene

- Oxidizing flame.

1. It contain more oxygen than any other flame.
2. This is used for oxidizing metal like brass,Zink etc.
3. It is similar to natural flame.

Arc welding

- In this process heat generation takes place by an electric arc.
- Both work piece and electrod is connected to different terminals of AC or DC su pply.
- small gap bet work piece and electrod arc is generated.
- It produces temp about 3600°c.
- It having following types.

1. Carbon Arc welding.
2. Shielded metal arc welding.
3. Gas shielded arc welding.
4. Plasma arc welding.

Carbon arc welding

- It is new welding process.
- Electric is made of carbon.
- Electrics are soft and not suitable for high temp.
- Electric consumption slowly & requires filler material.
- Arc is generated between electrode and w/p.
- DC supply is used Electrode is negative & w/p is positive.
- It is used for thick material.

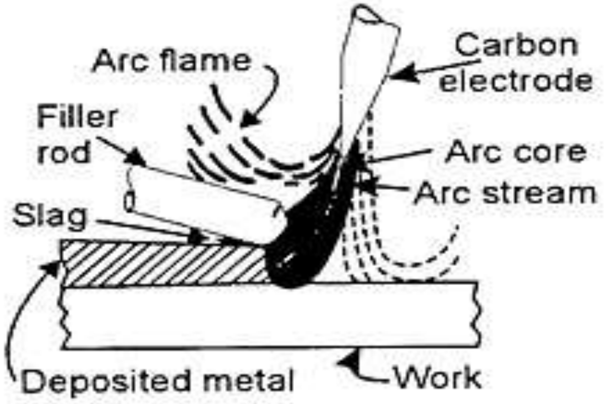
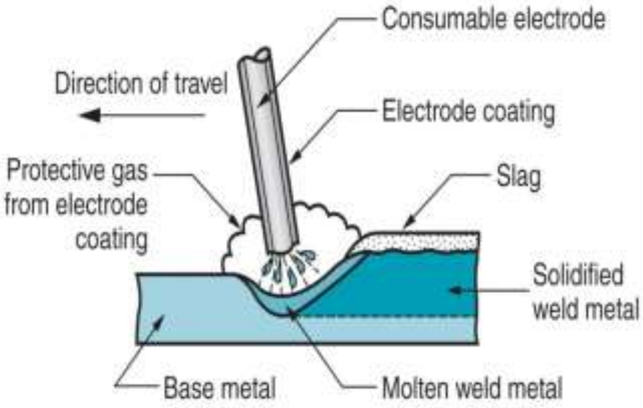


Fig. 15.2 Welding with the Carbon Arc

Shielded metal arc welding

- In this electric arc is used for the welding purpose.
- flux coated electrod is used for the welding.
- flux gives stability to protection to weld metal.
- during welding electrod is melt and serve as filler material.

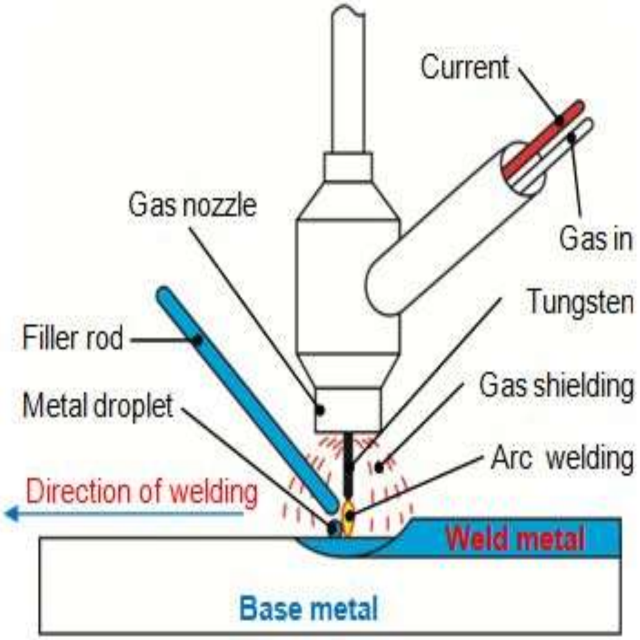
- flux over the electrode is also get melt to form coating over the weld material.A dvantages.
- Simple process.
- welding equipments are portable and less costly.
- Metal and alloy can be welded.
- Applicable in any position.



- Disadvantages.
- Automatic welding is difficult because of short electrod.
- Process is slow.
- Welding defects are more during replacement of electrod.

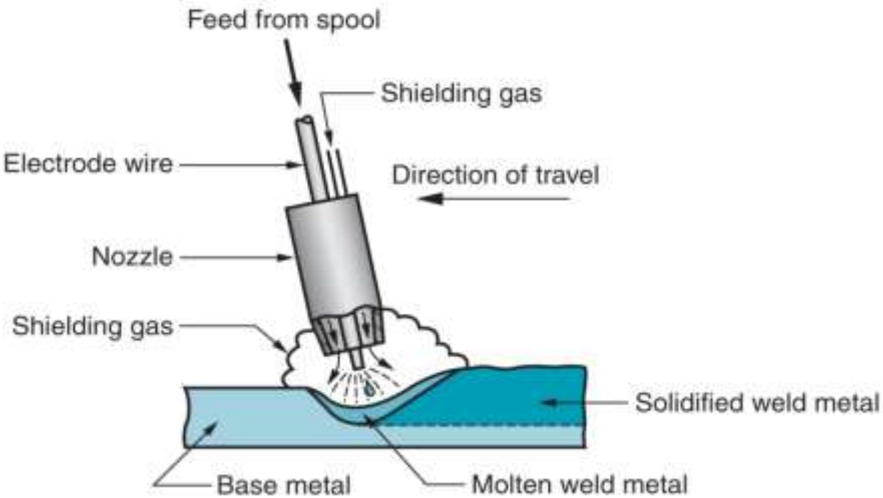
Tungsten inert gas welding(TIG)

- It is also called as gas tungsten arc welding.
- inert gas is act as shield and keep contaminant away from welded metal.
- Electrod does not melt during welding.it is non consumable electrod.
- filler material added separately.
- Inert gas used is argon.
- Advantages.
- Slag free operation due to absence of flux.
- different materials can be welded.
- no cleaning requires.
- Disadvantages.
- Equipment cost is high.
- slow process.
- Filler material requires



Metal inert gas welding (MIG)

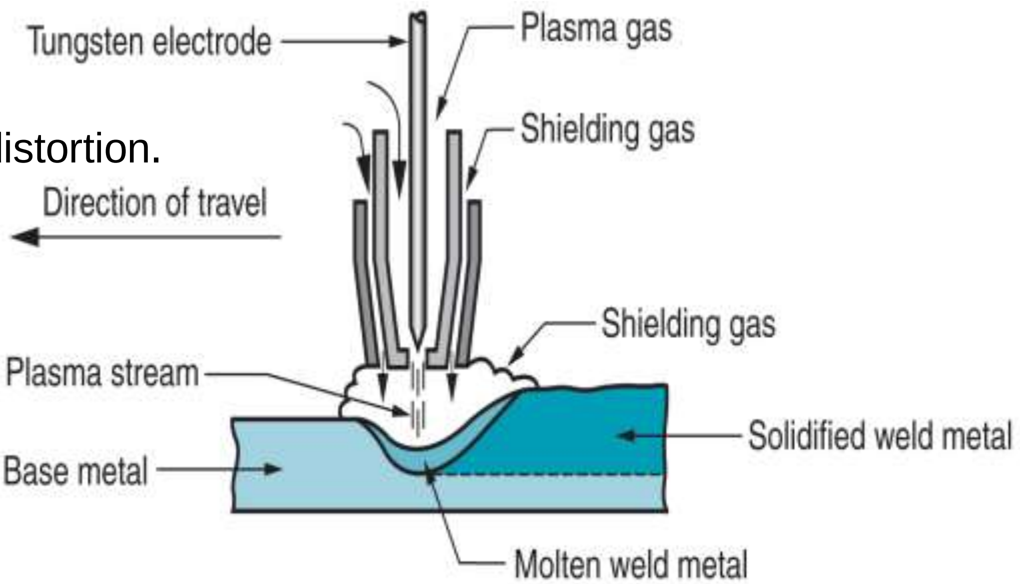
- It is also known as Gas metal arc welding (GMAW).
- In this process weld joint is produced by heating the metal with electric arc.
- The filling wire is continuously feed from wire reel at constant rate.
- wire is feed though the rollers at constant rate.
- Instead of flux ineart gas is supplied through welding torch.
- welding torch is connected to cylinder of unwary gas like argon
- Advantages.
- slag inclusion are not occurs.
- Continuous feed makes process faster.
- Less skilled operator is required.
- thick sheets can be welded.



- Disadvantages.
- Process is complicated.
- setup is complicated.
- Initial cost is high..
- Applications.
- For welding of commercial
- metals like Nickel,killed steel.

Plasma arc welding

- Plasma is heated ionized gas enable it to conduct electric current.
- Plasma arc is narrow restricted electric arc passes through water cooler orifice.
- It consist of non consumable tungsten electrod and shielding gas like argon.
- Argon gas flows through the orifice to form plasma.
- the temperature of plasma is about 10000°c.
- Advantages.
- Weld uniform throughout.
- greater depth of welding with less distortion.
- proper control on heat.
- metal disposition rate is high.
-



Plasma arc welding

- Limitations.
- Welding equipments are expensive.
- frequent replacement of nozzle.
- Applications
- Stainless steel welding.
- Nickel alloy in aero plane.
- Aluminium or graphite nozzles for rocket

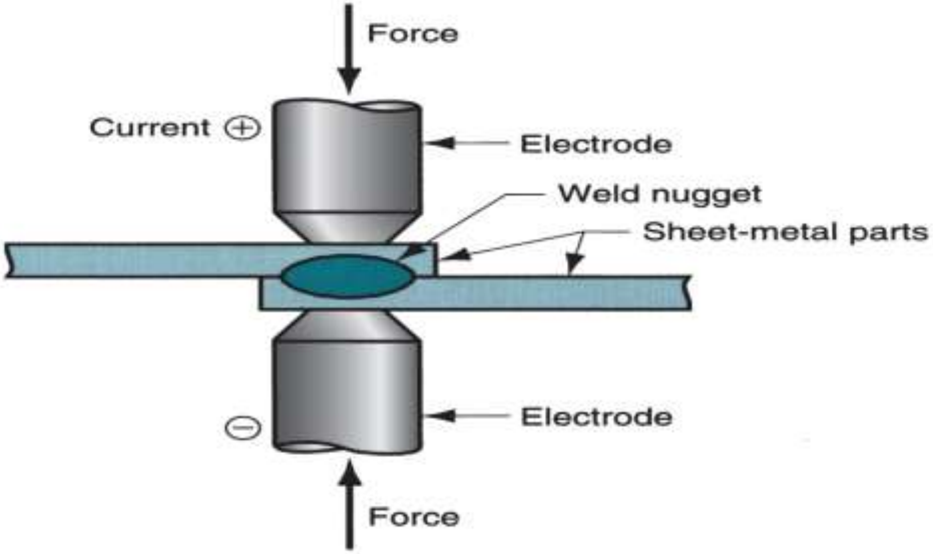
Resistance welding

- In these process heat and pressure is applied on joint.
- No additional filler or flux material required.
- joint is obtained by means of electrical resisitance.
- During process heavy current about 15000A is passed over joint with limited area for short time.
- Heavy current heat joint up to plastic state.
- due to high pressure metal get fused and forms joint.
- Heat generated is given as bellow H= CI2Rt
- It is classified as

1. Spot Welding
2. Seam Welding
3. Projection welding
4. Upset Welding

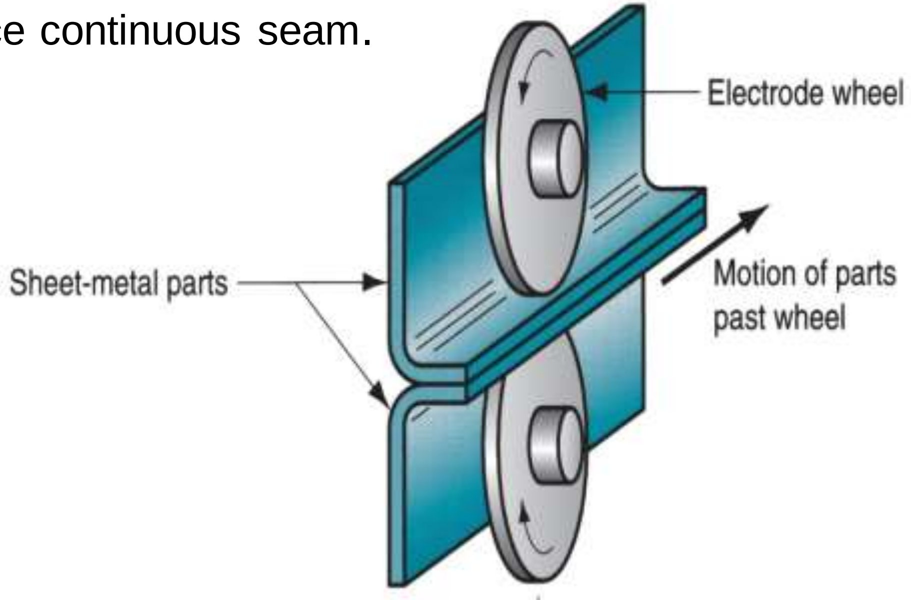
Spot Welding

- It is first type of resistance welding and widely used for lap joint of thin metal sheet.
- It is consist of two electrodes in which bottom electrode is fixed and upper is used for apply pressure.
- Sheets to be weld are kept in the two electrode .
- The pressure is applied over the sheets and current is passed though it so that they get weld.
- Welding is done due to high current is passed through the metal plates which produce heat.
- ## Applications
- For mass production.
- For Ferrous and Non ferrous metals.
- Used to weld sheets of 10mm thick



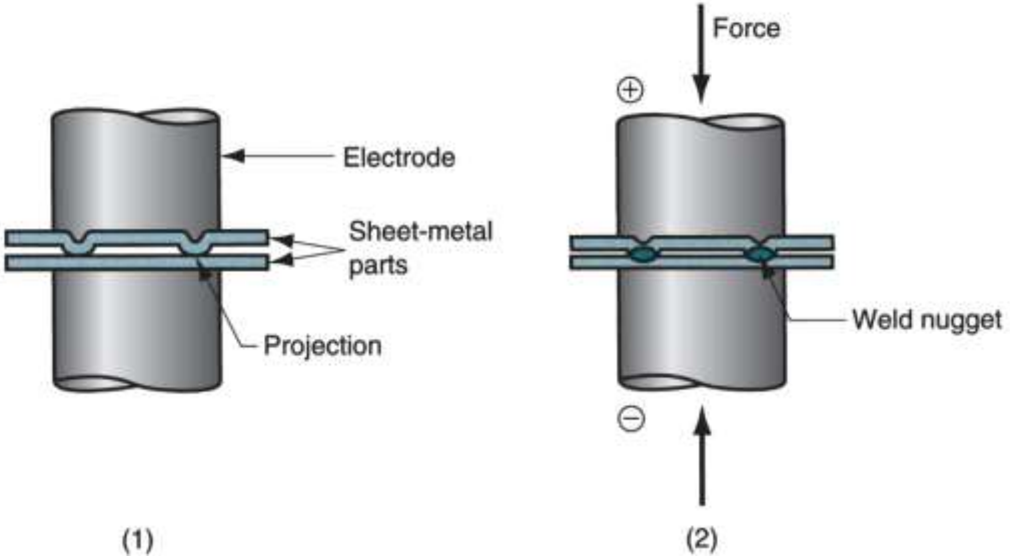
Seam Welding

- It is used for special purpose
- In this process disk type electrodes are used.
- It is used to produce continuous type of welding between two sheets.
- Disk are made up of cupper and current is applied through it
- Current is applied in pulse form with proper time interval
- Disks are rotate continuously so that metal sheets moves forward.
- Current can be regulated by timer to produce continuous seam.
- ## Applications.
- For gas or water tight joints
- To manufacture seam welded pipes



Projection Welding

- It is modification of spot welding.
- In this type one of the sheet is provided with projections and other is plane.
- These projections are in circular form.
- There diameter is equal to thickness of plate.
- Current and pressure is get concentrated at projection areas during process.
- During this process sheets to welded are kept between electrodes which made of cupper .
- Pressure is applied by upper movable electrode and lower is fixed .
- By passing current these projections are get melted and welding is done.



<u>For more detail contact us</u>